UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 23

Content.

1. (*Cont.*) Operator Norms/Singular Values of Random Matrices
2. Bounds in High Probability
3. Linear Regression

Theorem 1. *Let A bve an $m \times n$ random matrix with iid $\mathcal{N}(0,1)$ entries. Then,*

$$\sqrt{m} - \sqrt{n} \leq E\sigma_{\min}(A) \leq E\sigma_{\max}(A) \leq \sqrt{m} + \sqrt{n}$$

Proof of the Upper Bound.

$$\underbrace{\sigma_{\max}(A) = \|A\|}_{\text{Self-duality of Euclidean norm}} \overset{\text{Last Lecture}}{=} \sup_{u \in \mathbb{S}^{n-1}} \|Au\|.$$

Here, $\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n : \|x\|_2 = 1\}$. So,

$$E\sigma_{\max}(A) = E \sup_{(u,v) \in T} X_{uv},$$

where $X_{uv} = \langle Au, v \rangle$ and

$$T = \mathbb{S}^{n-1} \times \mathbb{S}^{m-1}$$

$$\langle Au, v \rangle = v^T Au = \sum_i v_i A_{ij} v_j$$

Bound the moment:

Fix $(u, v) \in T$ and $(u', v') \in T$. Control $E[(X_{uv} - X_{u'v'})^2]$. By direct calculation;

$$E\left[\left(\sum_{ij} v_i A_{ij} v_j - \sum_{ij} v' i A_{ij} v'_j\right)^2\right]$$

Notice $E[A_{ij}A_{i'k'}] = 0$ unless $\{i, j\} = \{i', j'\}$. Thus,

$$\begin{aligned}\sum_{ij}(v_i u_j - v'_i u'_j)^2 &= \sum_{ij}\left(v_i^2 u_j^2 - 2v_i v'_i u'_j u'_j + (v'_i)^2 (u'_j)^2\right) \\ &= \left(\sum_i v_i^2\right)\left(\sum_j v_j^2\right) - 2\left(\sum_i v_i v'_i\right)\left(\sum_j u)_j u'_j\right) \\ &\quad + \left(\sum_i (v'_i)^2\right)\left(\sum_j (u'_j)^2\right) = *\end{aligned}$$

so as

$$\|v_i\| = \|v'\| = \|u\| = \|u'\| = 1$$

we have

$$* = 2 - 2\langle v, v'\rangle_{\mathbb{R}^m}\langle u, u'\rangle\mathbb{R}^n.$$

A new process:

$$Y_{uv} := \langle g, u\rangle + \langle h, v\rangle$$

where $g \sim \mathcal{N}(0, I_n)$ and $h \sim \mathcal{N}(0, I_m)$ are independent.

$$\begin{aligned}E\left(Y_{uv} - Y_{u'v'}\right) &= E\left(\langle g, u - u'\rangle + \langle h, v - v'\right)^2 \\ &= E\langle g, u - u'\rangle^2 + 2\underbrace{E\langle g, u - u'\rangle\langle h, v - v'\rangle}_{=0 \text{ since } g \perp h} + E\langle h, v - v'\rangle^2\end{aligned}$$

Since $\langle g, u - u'\rangle \sim \mathcal{N}(0, \|u - u'\|^2)$ and $\langle h, v - v'\rangle \sim \mathcal{N}(0, \|v - v'\|^2)$, we get

$$E\left(Y_{uv} - Y_{u'v'}\right)^2 = \|u - u'\|^2 + \|v - v'\|^2.$$

Lemma 1. *For $u, u' \in \mathbb{S}^{n-1}$ and $v, v' \in \mathbb{S}^{m-1}$,*

$$2 - 2\langle u, u'\rangle\langle v, v'\rangle \leq \|u - u'\|^2 + \|v - v'\|^2$$

Proof of Lemma **??**.

$$\begin{aligned}\|u - u'\|^2 + \|v - v'\|^2 &= \|u\|^2 + \|u'\|^2 - 2\langle u, u'\rangle + \|v\|^2 + \|v'\|^2 - 2\langle v, v'\rangle \\ &= 4 - 2\langle u, u'\rangle - 2\langle v, v'\rangle\end{aligned}$$

So,

$$\|u-u'\|^2+\|v-v'\|^2-\big(2-2\langle u,u'\rangle\langle v,v'\rangle\big)\le 2+2\langle u,u'\rangle\langle v,v'\rangle-2\langle u,u'\rangle-2\langle v,v'\rangle$$
$$=2(1-\langle u,u'\rangle)(1-\langle v,v'\rangle) \qquad (*)$$

Since $\langle u,u'\rangle\le\|u\|\cdot\|u'\|=1$ (Cauchy-Schwarz) and similarly $\langle v,v'\rangle\le 1$, we have that $(*)$ is non negative. □

$$1+ab-a-b=(1-a)(1-b)$$

Therefore, X_{uv} and Y_{uv} are both centered Gaussian processes on T with

$$E\left(X_{uv}-X_{u'v'}\right)^2\le E\left(Y_{uv}-Y_{u'v'}\right)^2$$

Thus, by Sudakov-Fernique,

$$E\sigma_{\max}(A)=E\sup_{(u,v)\in T}X_{uv}\le E\sup_{(u,v)\in T}Y_{uv}=E\sup_{(u,v)\in T}\left(\langle g,u\rangle+\langle h,v\rangle\right).$$

The lat quantity is

$$=\underbrace{E\sup_{u\in\mathbb{S}^{n-1}}\langle g,u\rangle}_{E\|g\|(\mathcal{G}(B_2^n))}+\underbrace{E\sup_{v\in\mathbb{S}^{m-1}\langle h,v\rangle}}_{E\|h\|(\mathcal{G}(B_2^m))}$$

$$E\|g\|=E\sqrt{g_1^2+\cdots+g_n^2}\overset{\text{Jensen}}{\le}\sqrt{E\left(g_1^2+\cdots+g_n^2\right)}=\sqrt{n}$$

and similarly

$$E\|h\|\le\cdots\le\sqrt{m}.$$

□

Proof Sketch for Lower bound.

$$\sigma_{\min}(A)=\inf_{u\in\mathbb{S}^{n-1}}\|Au\|$$
$$\overset{\text{self duality}}{\le}\inf_{u\in\mathbb{S}^{n-1}}\sup_{v\in\mathbb{S}^{m-1}}\langle Au,v\rangle$$

Q: How to bound

$$E\inf_u\sup_v X_{uv}?$$

Tool: Gordon's ineq

Theorem 2 (Gordon)**.** *Let $(X_{ut})_{u\in U,t\in T}$ and $(Y_{ut})_{u\in U,t\in T}$ be two centered Gaussian process induced bh (u,t) in the product space $U\times T$. Suppose*

$$E\left(X_{ut}-X_{us}\right)^2 \leq E\left(Y_{ut}-Y_{us}\right)^2,\ \forall u,t,s$$

$$E\left(X_{ut}-X_{vs}\right)^2 \geq E\left(Y_{ut}-Y_{vs}\right)^2, \forall u\neq v,\ \forall t,s.$$

Then; for all $\tau\geq 0$:

$$\mathbb{P}\left(\inf_{u\in U}\sup_{t\in T} X_{ut}\geq\tau\right)\leq\mathbb{P}\left(\inf_{u\in U}\sup_{t\in T} Y_{ut}\geq\tau\right)$$

Thus,

$$E\left(\inf_{u\in U}\sup_{t\in T} X_{ut}\right)\leq E\left(\inf_{u\in U}\sup_{t\in T} Y_{ut}\right)$$

□

Gordon's inequality gives

$$E\sigma_{\min}(A)\geq\sqrt{m}-\sqrt{n}$$

Remark. *Substantial generalization of Gordon's inequalities were made by Hassibi-Oymak-Thrampoulidis.*

1 Bounds in High Probability

Theorem 3. *Let A be an $m\times n$ matrix wti h iid $\mathcal{N}(0,1)$ entries. Then for all $t>0$:*

$$\sqrt{m}-\sqrt{n}-t\leq\sigma_{\min}(A)\leq\sigma_{\max}(A)\leq\sqrt{m}+\sqrt{n}+t$$

with probability at least $1-2\exp\left(-\frac{t^2}{2}\right)$.

Theorem 4 (Concentration of Lipschitz Functions)**.** *Let $f:\mathbb{R}^n\to\mathbb{R}$ be a K-Lipschitz function with respect to Euclidean norm*

$$|f(x)-f(y)|\leq K\|x-y\|_{\mathbb{R}^n}, \forall x,y\in\mathbb{R}^n.$$

Then;

$$\mathbb{P}_{X\sim\mathcal{N}(0,I_N)}\left[|f(X)-Ef(X)|\geq t\right]\leq 2\exp\left(-\frac{t^2}{2K^2}\right)$$

Remark. *1. Proof can be found in Vershynin or Adler-Taylor*

2. *Also monograph by Ledoux-Talagrand.*

Proof of Upper Bound.

Claim 1. *Let $f : \mathbb{R}^{mn} \to \mathbb{R}$ be defined as $f(A) = \|A\|$. Then; f is 1-Lipschitz, i.e.,*

$$|f(A) - f(B)| \leq \|A - B\|_{\ell_2(\mathbb{R}^{mn}} = \|A - B\|_{\mathrm{F}} = \left(\sum_{i \leq n} \sum_{j \leq n} (A_{ij} - B_{i,j})^2 \right),$$

where $\| \cdot \|_F$ is the Frobenius norm. If the claim is true, then

$$\mathbb{P}\left(f(A) - Ef(A) \geq t\right) \leq \exp\left(-\frac{t^2}{2}\right)$$

So,

$$f(A) \leq Ef(A) + t \leq \sqrt{m} + \sqrt{n} + t$$

with probability at least $1 - \exp(-t^2/2)$. Lower bound is the same and a union bound gives $1 - 2\exp(-t^2/2)$.

Proof of Claim.

$$\|A\|_{\mathrm{op}} = \sup_{u \in \mathbb{S}^{n-1}, v \in \mathbb{S}^{n-1}} \langle Au, v \rangle$$

$$\begin{aligned} \langle Au, v \rangle - \langle Bu, v \rangle &= \langle (A - B)u, v \rangle \\ &\leq \sup_{u \in \mathbb{S}^{n-1}, v \in \mathbb{S}^{m-1}} \langle (A - B)u, v \rangle \\ &= \|A - B\|_{\mathrm{op}}. \end{aligned}$$

So, $\forall u, v$;

$$\begin{aligned} \langle Au, v \rangle &\leq \langle Bu, v \rangle + \|A - B\|_{\mathrm{op}} \\ &\leq \sup_{u \in \mathbb{S}^{n-1}, v \imath n \mathbb{S}^{m-1}} \langle Bu, v \rangle + \|A - B\|_{\mathrm{op}} \\ &= \|B\|_{\mathrm{op}} + \|A - B\|_{\mathrm{op}} \end{aligned}$$

Taking sup over (u, v) on the LHS:

$$\|A\|_{\mathrm{op}} - \|B\|_{\mathrm{op}} \leq \|A - B\|_{\mathrm{op}}$$

$$\|B\|_{\mathrm{op}} - \|A\|_{\mathrm{op}} \leq \|A - B\|_{\mathrm{op}}$$

□

Thus,

$$\begin{aligned}|f(A)-f(B)| &\le \|A-B\|_{\mathrm{op}} \\ &= \sigma_{\max}(A-B) \\ &\le \sqrt{\sum_i \sigma_i(A-B)^2} \\ &\overset{\text{definition of Frobenius norm}}{=} \|A-B\|_{\mathrm{F}} = \|A-B\|_{\ell^2(\mathbb{R}^{mn})}\end{aligned}$$

□

2 Linear Regression

Setup. Given an unknown $\beta^* \in \mathbb{R}^d$, define the measurements Y_i by

$$Y_i = \langle x_i, \beta^* \rangle + \varepsilon_i, i = 1, \cdots, n$$

where ε_i is the noise

Assumption 1. $\varepsilon_i \overset{indep}{\sim}$ *sub-Gaussian* (σ^2)

First Focus: Fixed design,

$$x_1, \cdots, x_n \in \mathbb{R}^d$$

are fixed. Given (x_i, Y_i), $i = 1, \cdots, n$, the goal is to "recover" the unknown β^*.

Modern literature: High dimensions, $n, d \to \infty$ but $n \ll d$.

Note that the measurement model corresponds to

$$Y = X\beta^* + \varepsilon$$

$$\underbrace{\begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}}_{\in \mathbb{R}^n} = \underbrace{\begin{bmatrix} \cdots x_1 \cdots \\ \vdots \\ \cdots x_n \cdots \end{bmatrix}}_{\in \mathbb{R}^{n \times d}} \underbrace{\begin{bmatrix} \beta^* \end{bmatrix}}_{\in \mathbb{R}^d} + \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n. \end{bmatrix}$$

Definition 1 (Orthonormal design)**.** *We call* $\{x_1, \cdots, x_n\}$ *an orthonormal design if* $\frac{1}{n} X^T X = I_d$*, where* I_d *is the identity matrix* $(d \times d)$.

Intuition: If $\{x_i\}$ is an orthonormal deisng, then $\|x\|^2 = n$,

$$\langle x_i, x_j \rangle = 0$$

This is "essentially" the case if X_is are sampled from a centered dist. with variance.

If X is an orthonormal design

$$\frac{1}{n}X^TY = \left(\frac{1}{n}X^TX\right)\beta^* + \frac{1}{n}X^T\varepsilon$$

Next lecture:

$$\tilde{Y} = \beta^* + \tilde{\varepsilon}$$

where $\tilde{\varepsilon}$ is still sub-Gaussian and β^* is Gaussian sequence model.